

Problem Background

In the current global energy transformation, new energy, as a substitute for petroleum-based energy sources, holds strategic importance in addressing climate crisis, carbon neutrality, and energy regeneration. As one of the new energy technologies, lithium battery technology has drawn industry attention to the reliability of battery products. Currently, factors considered to affect the lifespan and reliability of lithium batteries include: the number of charge-discharge cycles, temperature, depth of discharge, and different materials.

Problem Requirement

Based on the dataset provided in 'data1' and 'data2' and following the requirements for statistical case study writing, please address the following issues:

1. Based on the given information and data, including different battery components, four different accelerated temperature settings, 2,000 charge-discharge cycles, and the corresponding recorded remaining capacity of the battery cells, establish a nonlinear parametric statistical analysis model and perform parameter estimation. Predict the point estimate and interval estimate of the time when 50% of the battery products fail under high-temperature conditions of 50°C (a product is considered failed when its remaining capacity is less than 75%).
2. Based on the failure cycle counts of three different materials after charge-discharge cycles (data2), determine which material has better reliability and provide specific reasons.

Problem Analysis

The problem of modeling lithium battery reliability under varying conditions presents a multifaceted challenge that requires integrating statistical rigor with domain-specific insights. At its core, battery degradation is governed by complex electrochemical processes, where factors like temperature and charge-discharge cycles interact nonlinearly to accelerate capacity loss. A robust analysis must first acknowledge the physical underpinnings—such as the Arrhenius relationship for temperature-dependent degradation and the cumulative damage from cycling—before selecting appropriate statistical models. For instance, a Weibull or lognormal distribution might capture the time-to-failure distribution, but their suitability should be validated against empirical data, particularly for extrapolation to 50°C. Accelerated life testing models must account for potential deviations from Arrhenius behavior at high temperatures, where side reactions (e.g., electrolyte decomposition) could dominate, necessitating piecewise or physics-informed models. Parameter estimation should employ methods

like maximum likelihood with robust standard errors or Bayesian approaches to quantify uncertainty, especially when predicting beyond observed data ranges. Bootstrapping or Monte Carlo simulations can enhance interval estimates by incorporating both model and measurement uncertainties.

The comparison of material reliability demands a nuanced approach. Simple mean failure cycles are insufficient; instead, survival analysis tools like Kaplan-Meier curves or Cox proportional hazards models should be used to handle censored data and time-dependent covariates. Material performance may also exhibit batch effects or nonlinear degradation trajectories, suggesting the need for mixed-effects models or functional data analysis. For example, Material A might show slower initial degradation but a steep drop-off after a threshold, while Material B degrades linearly but with higher variability. Such patterns require tests beyond ANOVA, such as likelihood ratio tests for nested models or nonparametric comparisons of survival curves. The definition of "better reliability" should be context-specific—whether prioritizing median lifespan, 95th percentile failure time, or hazard rates under stress conditions—and explicitly stated to avoid ambiguity.

Interdisciplinary perspectives can further enrich the analysis. Incorporating electrochemical impedance spectroscopy data or differential voltage analysis could validate statistical models with mechanistic insights. Machine learning techniques, like Gaussian process regression or neural networks, might capture complex interactions missed by parametric models, though their interpretability trade-offs must be weighed. Real-world constraints—such as thermal management costs or material supply chain sustainability—should also inform the practical implications of the findings. For instance, a marginally superior material with high rare-earth dependency might be less viable than a slightly inferior but more sustainable alternative. Sensitivity analyses can explore how assumptions about failure thresholds (e.g., 75% vs. 80% capacity) impact conclusions, ensuring robustness across usage scenarios. By grounding statistical methods in domain knowledge and addressing both technical and practical dimensions, the analysis can deliver actionable insights for battery design and deployment in the energy transition.

The dynamic nature of battery aging also calls for iterative model refinement. Initial fits should be scrutinized for outliers or heteroscedasticity, which might indicate unmodeled factors like manufacturing defects or uneven stress distributions. Cross-validation can assess predictive performance, while residual analysis can identify systematic biases, prompting model adjustments—for example, adding random effects for batch variability or interaction terms for temperature-cycling synergies. Open questions, such as the impact of partial charge cycles or rest periods between cycles, could be explored through controlled experiments or latent variable modeling. Ultimately, the analysis must balance complexity and generalizability, providing a framework adaptable to evolving battery technologies and usage patterns. This holistic approach ensures that the model not only fits historical data but also remains predictive under novel conditions, guiding strategic decisions in new energy adoption.

Tasks

Task 1

Task Description

Subtask 1: Data Preparation and Exploratory Analysis

This subtask involves systematically processing and analyzing the raw datasets ('data1' and 'data2') to ensure data quality and uncover preliminary insights. For 'data1', which contains battery capacity measurements across multiple temperatures, charge-discharge cycles, and materials, the first step is data cleaning: handling missing values (e.g., interpolation or removal), identifying and addressing outliers (e.g., using Tukey's fences or domain-specific thresholds), and standardizing capacity measurements by normalizing to initial values to account for batch variability. Exploratory Data Analysis (EDA) is then performed using visualization tools (e.g., scatterplots for degradation trends, boxplots for cycle-wise capacity distribution, and heatmaps for temperature effects) and statistical summaries (e.g., mean capacity loss per temperature group, variance decomposition). Key relationships, such as nonlinear capacity decay over cycles or temperature-accelerated degradation, are identified using smoothing techniques (e.g., LOESS curves) or correlation analysis. For 'data2', which records failure cycles for three materials, data integrity is verified (e.g., checking for censored or incomplete entries), and materials are categorized for comparison. Descriptive statistics (e.g., median failure cycles, interquartile ranges) and nonparametric survival analysis (Kaplan-Meier curves) are generated to visualize material performance differences. Feature engineering may include deriving time-dependent covariates (e.g., cumulative damage) or transforming temperature into Arrhenius terms ($1/kT$) for later modeling. Tools like Python (Pandas, NumPy, Matplotlib/Seaborn) or R (tidyverse, survival) are employed for automation and reproducibility. The output is a cleaned, annotated dataset with documented metadata (e.g., outlier handling rules) and EDA reports (visualizations and summary tables) to guide subsequent modeling. This stage ensures data readiness and informs the choice of parametric forms or hypotheses for Subtask 2.

Task Analysis

The core objective of Subtask 1 is to systematically prepare and analyze the raw datasets ('data1' and 'data2') to ensure data quality and uncover preliminary insights, which will serve as the foundation for subsequent modeling efforts. This task is critical within the larger project as it directly influences the reliability of downstream analyses, such as predictive modeling of battery degradation or material failure, by addressing data inconsistencies, identifying patterns, and generating actionable summaries. The significance lies in its role in transforming raw, potentially noisy data into a structured, interpretable form, enabling robust hypothesis testing and model selection in later stages. Key